

D. Not Alone

time limit per test: 2 seconds
 memory limit per test: 256 megabytes

A circular array b of length m is *nice* if every element has at least one adjacent element* that is equal to it. Formally, for every $1 \leq i \leq m$, at least one of the following holds:

$b_i = b_{(i+m-2) \bmod m + 1}$, or $b_i = b_{i \bmod m + 1}$, where $x \bmod y$ denotes the remainder from dividing x by y .

You are given a circular array a of length n . In one operation, you may increase or decrease any element of a by 1. Your task is to determine the minimum number of operations required to make array a nice. More formally, find the minimum value of $\sum_{i=1}^n |b_i - a_i|$ among all nice circular arrays b of length n .

* In a circular array of length m :

- For each index $2 \leq i \leq m - 1$, the element at index i is adjacent to the elements at indices $i - 1$ and $i + 1$.
- The element at index 1 is adjacent to the elements at indices 2 and m .
- The element at index m is adjacent to the elements at indices $m - 1$ and 1.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains a single integer n ($3 \leq n \leq 2 \cdot 10^5$) — the length of the circular array a .

The second line of each test case contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$) — the elements of the circular array a .

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer representing the minimum number of operations required to make array a nice.

Example

input	Copy
<pre>4 5 1 1 1 1 4 2 100 99 3 5 2 2 5 9 5 6 1 1 1 2 1 2</pre>	
output	Copy
<pre>0 2 4 1</pre>	

Note

In the first test case, all elements of a are equal. Therefore, the circular array a is already nice and no operations are required.

In the second test case, we can perform the following sequence of operations:

Codeforces Round 1057 (Div. 2)

Contest is running

01:25:05

Contestant



→ Submit?

Language: Python 3.13.2
 Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

Submit

→ Score table

	Score
Problem A	432
Problem B	648
Problem C	1080
Problem D	1512
Problem E	1944
Problem F	2808
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 00:34 from the first attempt

- Increase a_1 by 1. Now, $a = [3, 100, 99, 3]$.
- Decrease a_2 by 1. Now, $a = [3, 99, 99, 3]$.

After these operations, every element has at least one adjacent element with the same value:

- a_1 is equal to a_4 .
- a_2 is equal to a_3 .
- a_3 is equal to a_2 .
- a_4 is equal to a_1 .

In the third test case, the circular array a can become nice by decreasing a_4 four times. This results in $a = [2, 2, 5, 5]$ which is nice as every element has at least one adjacent element with the same value.

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